

Official Examination Answer Key

Question 1

Question 1: Explain the primary difference between a list and a tuple in Python.

Answer Key: A list is mutable, meaning its elements can be changed, added, or removed after creation, and it is defined using square brackets []. A tuple is immutable, meaning its contents cannot be altered once defined, and it uses parentheses ().

Question 2

Question 2: What is the purpose of the 'init' method in a Python class?

Answer Key: The `__init__` method is a constructor function that runs automatically when a new object of a class is instantiated. Its main purpose is to initialize the object's attributes with initial values.

Question 3

Question 3: Define the concept of Overfitting in Machine Learning.

Answer Key: Overfitting occurs when a machine learning model learns the training data too well, capturing the noise and fluctuations instead of the underlying pattern. This causes high accuracy on training data but poor generalization on unseen test data.

Question 4

Question 4: What does the acronym SQL stand for and what is its primary use?

Answer Key: SQL stands for Structured Query Language. It is a standard

programming language specifically designed for managing, updating, querying, and manipulating data stored in relational databases.